Family Clinic Platform

Three dashboards, one secure system: Family Clinic, Medical Aid Care Plans and the Next.js invoice dashboard

Built by Ammara Badat | Next.js, Supabase and Vercel

The problem we solve

A family practice runs on separate tools: bookings, records, payments, plans and invoices.

Scattered

Patients, doctors, reception and pharmacy each need different information

Risky

Sensitive health data must only reach the right person

Slow

Booking, paying and getting an invoice should take minutes, not calls

One platform, three dashboards

The landing page opens each one from its own card.

Family Clinic

Patients, doctors, reception, pharmacy and stock

Next.js invoices

Customers, invoices and revenue (the original dashboard)

Medical Aid

Care Plans: join, pay by a chosen card, see cover and PDFs

Family Clinic: four doors

Each door has its own sign-in and shows only what that role needs.

Patients

/medi-clinic

Admin

/medi-clinic/dashboard

Doctors

/medi-clinic/doctors

Pharmacy

/medi-clinic/pharmacy

The doctor experience

Sign in, choose who you are, and the dashboard is yours.

Pop-up

"Which doctor are you?" lists the Dr. Sample doctors

Their dashboard

Today, coming up, patients and weekly shifts

Mon to Fri

Shifts are set as a day range and shown in a weekly grid

Calendar

A week grid with available hours shaded and hover details

The patient journey

Everything a family needs after one sign-up.

Book

Pick a provider and an open slot

Upload

Documents go to a private bucket

Manage

Edit your own profile, save cards, join a plan

Track

Appointments, prescriptions, claims, invoices

Medical Aid and Care Plans

Enquire in the portal, finish in the Care Plans dashboard.

1. Choose

Essential R199, Family Plus R549 or Premium R899 a month

2. Sign up

Same accounts as the clinic

3. Pay

Pick which saved card is debited (demo payment)

4. See cover

Plan, payments and sample PDFs appear

Plan progress and claims

Members see where they are; staff keep claims moving.

Progress

12-month bar, next payment date, total paid, payment schedule

Claims

Submit, then follow Submitted, In review, Approved or Rejected, Paid

Staff and export

Staff move each claim one step; members page and CSV export

Database rules

Nobody can jump a claim straight to paid

Run the clinic safely

Tools for the owner, admin and stock room.

Audit log

Who changed what, and when; no personal values copied

Stock alerts

Low stock, expiring in 30 days, expired

No double booking

The database refuses a clash

Members

Care Plan members, with a Payment due badge

Works while you sleep

Daily jobs and an installable app.

Reminders

E-mail and SMS before an appointment, once each

Renewals

Due plans renew; unusable cards are flagged

Installable

Add to the home screen; offline page only, nothing stored

Invoices created automatically

A PDF invoice exists for every payment and every appointment with services.

Care Plan payments

Download PDF from the payments table

Clinic payments

Link in the portal and staff register

Appointments

Built from recorded services; Unpaid until paid

Easy to move around

Small upgrades that make it feel professional.

Arrows

Back, forward and refresh on every login page, hub and dashboard

Clear layout

Header and tab bar by role, loading skeletons between pages

Polish

Branded 404, favicon, smooth scrolling, visible focus, tidy numbers

Make it yours

A toolbar in every header, on every page.

Dark and light

One button, remembered, no flash on reload

English and Afrikaans

Navigation, headers and key pages in both languages

Keyboard shortcuts

T theme, L language, G then a letter to jump; ? lists the keys

Live and guided

The platform updates itself and explains itself.

Live updates

Pages refresh by themselves when data changes

Notification bell

Unread count for appointments, payments and claims

Guided tour

Ten plain-language steps, in both languages

Security and honesty

Least privilege everywhere, and no overclaiming.

Trusted roles

Set in Supabase Auth, not editable by users

Row-level security

The database blocks other patients' data

Demo only

No payment gateway; only brand, last four digits and expiry stored

Claim rules

Allowed stage moves are checked by the database

How it is built and shipped

Modern, simple and repeatable.

Next.js + TypeScript

Server pages and actions keep secrets on the server

Supabase

Auth, Postgres, row-level security and private storage

Vercel from main

Merging to main publishes the site

Proof it works

Checked automatically on every push and after every deploy.

Security tests

Row-level security tested on a real PostgreSQL

Build and types

GitHub builds and type-checks the app

Browser tests

The live site is tested after each deploy

Lighthouse

Accessibility 96 to 100, performance 99 to 100

What comes next

The honest roadmap.

Payments

Connect the PayFast sandbox checkout (webhook built)

Refunds

Refunds for plan members

Language

Translate the remaining pages into Afrikaans

Compliance

Legal and privacy review before real patient data

Speaker notes (about 6 minutes)

1. Cover (8s) Say the one-liner: three dashboards in one secure system. Name yourself and the tools.

2. The problem (20s) A practice juggles bookings, records, payments, plans and invoices. Health data is sensitive, so each role must see only what it needs.

3. Three dashboards (20s) Open the landing page. Point at the three cards and logos: clinic blue, Next.js navy, Medical Aid teal. Each card opens its own dashboard.

4. Four doors (20s) Patients, admin, doctors, pharmacy. Same app, different sign-in and different view. Mention roles live in trusted Auth metadata.

5. Doctor experience (30s) Sign in as a doctor. The pop-up asks which doctor. Show today, coming up and the Monday to Friday shifts, then open Calendar and hover an appointment. Use the header Switch button last. Only the Dr. Sample providers are used.

6. Patient journey (30s) Book an appointment, upload a document, save a card, then open My profile and edit the phone number. Everything is the patient's own data.

7. Care Plans (30s) From the portal press Enquire about a plan. Sign in, join, choose the card. Stress: demo payment, no money moves, no card number or CVV stored.

8. Plan progress and claims (40s) Show the progress bar and payment schedule. Open the Claims tab and submit a claim. On a staff account move it one step on the Claims board. Say clearly: claims are tracked by hand, nothing is adjudicated automatically.

9. Run the clinic safely (30s) As owner, open Audit log and show who changed what and when. On the pharmacy page show the stock alerts. Mention the database refuses a double booking and the Members page shows a Payment due badge.

10. Works while you sleep (25s) Daily jobs send reminders by e-mail and SMS and renew Care Plans. The site can be installed on a phone and shows only an offline page when disconnected; it never stores records.

11. Invoices (20s) Open an invoice PDF from the payments table. Numbers are fixed per record. Appointment invoices appear once services are recorded.

Speaker notes (continued)

12. Navigation and polish (20s) Show the back, forward and refresh arrows on a login page and inside a dashboard. Mention the 404 page and loading skeletons.

13. Make it yours (30s) Toggle dark mode, switch to Afrikaans, press ? to show the shortcut list, then G and P to jump to Patients.

14. Live and guided (30s) Keep a staff window and a patient window side by side. Submit a claim in one and watch the bell in the other. Then press the sparkle button to start the guided tour.

15. Security and honesty (25s) Row-level security, least privilege, private storage, rules enforced in the database. Be upfront: demo payments only, legal review needed before real data, no compliance claim.

16. Build and shipping (15s) Next.js on Vercel, Supabase behind it. Merge to main and Vercel publishes.

17. Proof it works (20s) GitHub runs the security tests on a real Postgres, builds the app, then browser tests and Lighthouse check the live site. Accessibility 96 to 100, performance 99 to 100.

18. Next steps (15s) Connecting the PayFast sandbox checkout, refunds, finishing the Afrikaans translation, and a legal and privacy review. Then invite questions.

Likely questions and short answers

Q: Is the payment real?
No. Joining a plan records a clearly labelled demo payment. No gateway is connected and no money moves.

Q: Where is card data kept?
Only brand, last four digits and expiry. Full numbers and CVV are never stored.

Q: Can a patient see another patient's data?
No. Row-level security in the database blocks it, even if someone edits the page.

Q: Why three dashboards?
They serve different audiences: clinic operations, billing and plans. One repository and one login system keep it simple.

Q: How are invoices created?
On request from stored payments and appointment services, in one PDF layout with a number based on the record.

Q: Is it compliant with POPIA or HIPAA?
No certification is claimed. The design follows least privilege and minimal data, and legal review is planned before real data.

Q: Can I use it in Afrikaans?
Yes, with the language button. Navigation, headers, calendar, claims, profile, notifications and tour are translated; other page text is still English.

Q: How do live updates work?
Supabase Realtime tells the app when data changes; the page refreshes and the bell shows a notification. Access rules still decide who receives what.

Q: Are claims approved automatically?
No. Staff move each claim through the stages by hand, and the database only allows valid moves.

Likely questions (continued)

Q: How do you know it works?
Every push runs the security tests on a real PostgreSQL, a build and type check, then browser tests and Lighthouse on the live site.

Q: Who changed this record?
The Audit log shows who, when and which fields changed, without copying personal values.

Q: How does a change go live?
Merge to the main branch; Vercel builds and publishes it. A failed build leaves the previous version live.

Live demo order (9 minutes)

1. Open the landing page and show the three cards and logos.
2. Family Clinic card, then sign in as a doctor: pick a Dr. Sample doctor, show today and the weekly shifts, open Calendar, then use Switch.
3. Sign in as a patient: book an appointment, open Documents and Cards, then edit the phone number under My profile.
4. Press Enquire about a plan: land in Care Plans, join a plan, choose a card, then show the progress bar.
5. Open the Claims tab, submit a claim, then move it forward from a staff account on the Claims board and watch the bell.
6. Open the Payments tab and download the invoice PDF.
7. Toggle dark mode and Afrikaans, press ? for shortcuts, and start the tour.
8. As owner, open Audit log and the stock alerts on the pharmacy page.
9. Show the arrows on a login page, then the 404 page. Finish on the Next.js card and take questions.